

FORMALASR: END-TO-END SPOKEN CHINESE TO FORMAL TEXT

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ABSTRACT

Automatic speech recognition (ASR) systems are typically optimized for verbatim transcription, which preserves disfluencies, filler words, and informal spoken structures that are often unsuitable for downstream writing-oriented applications. A common workaround is a two-stage ASR+LLM pipeline for post-editing, but this design increases latency and memory cost and is difficult to deploy on-device. We present FormalASR, two compact end-to-end models (0.6B and 1.7B) that directly transcribe spoken Chinese into formal written text. To enable this setting, we build WenetSpeech-Formal and Speechio-Formal, two large-scale spoken-to-formal datasets constructed by LLM-based rewriting and quality filtering. We then fine-tune Qwen3-ASR at two scales (0.6B and 1.7B) with supervised fine-tuning. Experiments on WenetSpeech-Formal and Speechio-Formal show that FormalASR achieves up to 37.4% relative CER reduction over verbatim baselines, while also improving ROUGE-L and BERTScore. FormalASR requires no post-processing LLM at deployment time, providing a lightweight, on-device solution for spoken-to-formal transcription.

Index Terms— speech recognition, spoken-to-formal, on-device ASR, text formalization, supervised fine-tuning

1. INTRODUCTION

Automatic speech recognition (ASR) has become a foundational component of modern human-computer interaction, powering applications ranging from voice assistants and meeting transcription to real-time captioning and document dictation. State-of-the-art systems such as Whisper [1], Qwen3-ASR [2], and SenseVoice [3] have achieved remarkable accuracy on standard benchmarks, yet they share a fundamental design assumption: the output should faithfully reproduce the spoken surface form. This verbatim transcription paradigm faithfully captures what was said, but the output inherits all the characteristics of spontaneous speech—filler

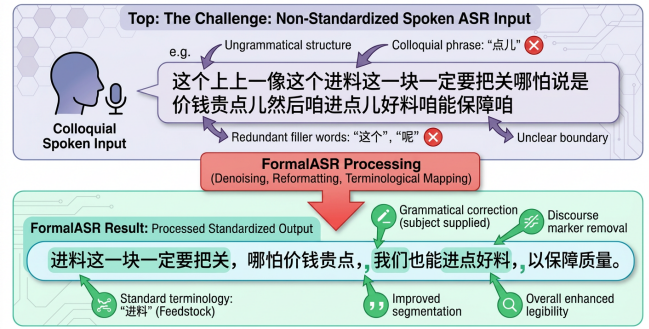


Fig. 1. An example of spoken-to-formal conversion. The verbatim ASR transcript preserves disfluencies and informal spoken patterns, while FormalASR directly produces a clean, formal written sentence from the same audio input.

words, false starts [4, 5], repetitions, and loosely structured sentences [6] that would be unacceptable in formal documents. In practice, downstream consumers of ASR output—meeting minutes generators, dialogue systems, voice-controlled document editors—expect formal, well-formed text, not a verbatim transcript of how someone actually spoke.

A common remedy is a two-stage pipeline: first transcribe verbatim with an ASR model, then apply a separate large language model (LLM) to rewrite the transcript into formal style [7]. While effective in server-side settings, this approach doubles memory footprint and inference latency, allows errors to propagate across stages, and—most critically—makes the system unsuitable for on-device or privacy-sensitive deployments where only a compact model can run. Large multimodal models such as GPT-4o-audio-preview [8] can produce formal-style transcriptions in a single pass, but depend on cloud APIs, incurring per-token costs and raising privacy concerns that preclude edge deployment. As illustrated in Figure 1, the gap between verbatim ASR output and the expected formal written form can be substantial even for a single utterance.

In this work, we propose FormalASR, an end-to-end ap-

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proach that directly maps spoken Chinese audio to formal written text using a single compact model with no auxiliary LLM required at inference time. The key insight is that an audio-language model can be taught to perform acoustic recognition and linguistic formalization *simultaneously*, provided it is trained on appropriate spoken-to-formal supervision. To supply this supervision at scale, we construct WenetSpeech-Formal and Speechio-Formal, two large-scale Chinese spoken-to-formal ASR datasets derived from WenetSpeech [9] and Speechio [10], built by rewriting verbatim transcriptions with DeepSeek-V3.2 [11] and applying quality filtering. We then fine-tune Qwen3-ASR in two scales, 0.6B and 1.7B, on these datasets using supervised fine-tuning (SFT). Our main contributions are:

- We construct and open-source WenetSpeech-Formal¹ and Speechio-Formal², two large-scale Chinese spoken-to-formal ASR datasets built by rewriting verbatim transcriptions with an LLM and applying quality filtering, enabling end-to-end spoken-to-formal training without any auxiliary model at inference time.
- We propose and open-source FormalASR, compact audio-language models in two scales (0.6B³ and 1.7B⁴), fine-tuned with SFT, achieving up to 31.4% relative CER reduction and consistent ROUGE-L and BERTScore improvements over the verbatim baseline on both in-domain and cross-domain Chinese benchmarks, while remaining suitable for on-device deployment.

To the best of our knowledge, FormalASR is the first work to fine-tune a compact audio-language model end-to-end for spoken-to-formal Chinese transcription. Our results reveal that modern ASR models already possess the latent capacity for linguistic formalization—they simply need appropriate supervision to activate it, without any increase in model size or inference-time complexity.

2. RELATED WORKS

2.1. Automatic Speech Recognition

Modern ASR systems have evolved from traditional hybrid HMM-DNN architectures [12] toward end-to-end models based on CTC [13] and attention-based encoder-decoder frameworks [14]. Large-scale pre-trained models have further advanced the field: Whisper [1] demonstrates that training on hundreds of thousands of hours of weakly supervised audio data yields robust multilingual transcription, while audio-language models such as Qwen3-ASR [2] and SenseVoice [3]

integrate a powerful language model decoder to improve recognition of rare words and domain-specific terminology. Despite these advances, all of these systems are designed to produce *verbatim* transcriptions that faithfully preserve the spoken surface form, including filler words such as “um”, “uh”, and “you know”, false starts, and informal sentence structures. This design choice is appropriate for transcription benchmarks measured by Character Error Rate (CER), but it means that the output is not directly suitable for downstream applications such as document generation, dialogue systems, or voice-controlled interfaces that expect formal output.

2.2. Speech-to-Text Formalization

Converting spoken-style transcriptions into formal text is a long-standing challenge. Early approaches rely on hand-crafted rules or finite-state transducers to handle specific surface phenomena such as number verbalization and punctuation insertion [15, 16], but these methods cannot handle the broader linguistic restructuring required for spoken-to-formal conversion. Disfluency detection methods [4, 5] identify and remove filler words and false starts, yet do not address deeper structural formalization such as sentence reorganization or register conversion [6]. A more flexible paradigm chains a compact ASR model with a large language model that converts verbatim output into formal text [7], leveraging the strong language understanding of modern LLMs to handle diverse spoken-language phenomena; however, this requires two models to be loaded simultaneously, doubling memory footprint and inference latency, and makes on-device deployment infeasible. Large multimodal models such as GPT-4o-audio-preview [8] can produce fluent, formal-style transcriptions from raw audio in a single forward pass, but depend on cloud APIs, incurring per-token costs and raising privacy concerns that preclude on-device or latency-sensitive deployment. As summarized in Table 1, no existing approach simultaneously achieves spoken-to-formal conversion, on-device deployability, single-model inference, and low cost.

FormalASR addresses this gap by fine-tuning 0.6B and 1.7B compact audio-language models to directly produce formal output from speech in a single forward pass, requiring no auxiliary model at inference time and remaining suitable for on-device deployment.

3. DATASETS: WENETSPEECH-FORMAL AND SPEECHIO-FORMAL

3.1. Construction Pipeline

We construct WenetSpeech-Formal and Speechio-Formal from the WenetSpeech corpus [9] and Speechio benchmark data [10], following a three-stage pipeline:

Verbatim transcription collection. We use the original audio files and their verbatim transcriptions from WenetSpeech and Speechio as input. These verbatim transcripts preserve spoken-language characteristics including filler words,

¹WenetSpeech-Formal: <https://huggingface.co/datasets/TaurenMountain/WenetSpeech-Formal>

²Speechio-Formal: <https://huggingface.co/datasets/TaurenMountain/Speechio-Formal>

³FormalASR-0.6B: <https://huggingface.co/TaurenMountain/FormalASR-0.6B>

⁴FormalASR-1.7B: <https://huggingface.co/TaurenMountain/FormalASR-1.7B>

Table 1. Comparison of speech-to-text paradigms. FormalASR achieves spoken-to-formal conversion while remaining on-device deployable, single-model, and low-cost.

Paradigm	Spoken-to-formal	On-device capable	Single model	Low cost
Traditional ASR	×	✓	✓	✓
ASR + LLM	✓	×	×	×
Multimodal LLM	✓	×	✓	×
FormalASR (Ours)	✓	✓	✓	✓

false starts, repetitions, and informal sentence structures.

LLM-based formalization. We prompt DeepSeek-V3.2 [11] to rewrite each verbatim transcript into formal written Chinese. The rewriting process includes removing filler words and disfluencies, restructuring sentences to follow written conventions, normalizing punctuation and spacing, and correcting obvious recognition errors while preserving semantic content. The prompt instructs the model to produce concise, grammatically correct written text that conveys the same meaning as the spoken input.

Quality filtering. We apply automatic filtering to discard low-quality rewrites. Samples are removed if the rewritten text is semantically inconsistent with the original as measured by embedding similarity, if the edit distance between verbatim and formal text is too small to indicate meaningful formalization or too large suggesting potential hallucination, or if the rewritten text contains obvious errors or artifacts. This filtering step ensures that the training data contains only high-quality spoken-to-formal pairs.

3.2. Dataset Details

Table 2 summarizes the statistics of WenetSpeech-Formal and Speechio-Formal. WenetSpeech-Formal contains 969K training samples derived from the WenetSpeech corpus, covering diverse domains including audiobooks, podcasts, news broadcasts, and conversational speech. Speechio-Formal consists of 43K test samples spanning 27 domain-specific subsets (ZH00000–ZH00026), including lecture recordings, interviews, meeting transcripts, and spontaneous dialogue, providing a comprehensive cross-domain evaluation benchmark.

Each sample consists of an audio file paired with two text fields: `original_text` (verbatim transcription) and `target_text` (formal written text produced by DeepSeek-V3.2). Table 3 shows representative examples illustrating the types of formalization performed, including filler-word removal, error correction, and sentence restructuring.

By training on these spoken-to-formal pairs, FormalASR learns to directly produce formal written output from speech, eliminating the need for a separate post-processing stage.

Table 2. WenetSpeech-Formal and Speechio-Formal dataset statistics.

Dataset	# Samples	Usage
WenetSpeech-Formal (train)	969,201	SFT training
WenetSpeech-Formal (test)	31,932	In-domain eval
Speechio-Formal (test)	43,178	Cross-domain eval

Table 3. Representative spoken-to-formal conversion examples.

Verbatim (original_text)	Formal (target_text)
把这个呃增加的这个利润	把这个增加的利润。
对全美国全球影响影响不大	对美国全球影响不大。
但是我想这里这当中就是如果如果一定要那个挑一点儿什么的话	但是，如果一定要从挑出一点什么的话。

4. METHOD

Given an input audio utterance $\mathbf{x}$, our objective is to directly predict a formal written transcription $\hat{y}$ in a single pass:

$$\hat{y} = \arg \max_y P_\theta(y | \mathbf{x}), \quad (1)$$

where y denotes a well-formed written sentence rather than a verbatim spoken transcript. Different from the conventional ASR→LLM pipeline, this formulation couples acoustic recognition and linguistic formalization into one conditional generation process, so no auxiliary rewriter is required at inference time.

We instantiate $P_\theta(y | \mathbf{x})$ with Qwen3-ASR [2] (0.6B and 1.7B variants), which adopts a Whisper-style audio encoder and an autoregressive Qwen decoder. For each training sample, the model receives speech features extracted from the waveform and is supervised to generate the corresponding formal target text from WenetSpeech-Formal and Speechio-Formal. The training objective is standard teacher-forced maximum likelihood:

$$\mathcal{L}_{\text{SFT}} = - \sum_{t=1}^T \log P_\theta(y_t | \mathbf{x}, y_{<t}), \quad (2)$$

where T is the target length and y_t is the t -th token in the formal reference. Optimizing this objective encourages the model to jointly learn: acoustic-to-text alignment, disfluency removal, spoken-to-formal style transfer, and content-preserving rewriting under a unified end-to-end framework⁵.

During inference, decoding is performed directly on audio input to produce formal text, which keeps system complexity identical to a standard single-model ASR deployment while delivering spoken-to-formal outputs.

⁵Our code is available at <https://github.com/TaurenMountain/FormalASR>

Table 4. Spoken-to-formal ASR results on WenetSpeech-Formal and Speechio-Formal benchmarks.

Model	WenetSpeech-Formal			Speechio-Formal		
	CER ↓	ROUGE-L ↑	BERTScore ↑	CER ↓	ROUGE-L ↑	BERTScore ↑
Qwen3-ASR-0.6B	0.2581	0.8463	0.9198	0.2252	0.8701	0.9343
FormalASR-0.6B (Ours)	0.1770	0.8769	0.9359	0.1603	0.8948	0.9481
Qwen3-ASR-1.7B	0.2460	0.8571	0.9268	0.2393	0.8510	0.9108
FormalASR-1.7B (Ours)	0.1606	0.8896	0.9439	0.1499	0.9029	0.9533
Whisper large-v3	0.3631	0.7393	0.8538	0.3302	0.7643	0.8795

5. EXPERIMENTS

5.1. Experimental Setup

We fine-tune Qwen3-ASR at two scales (0.6B and 1.7B) on WenetSpeech-Formal using full-parameter supervised fine-tuning (SFT). Both models are initialized from the official Qwen3-ASR [2] checkpoints and trained for 2 epochs on the 969K-sample training split. All experiments are conducted on 2 NVIDIA A800-SXM4-80GB GPUs. Training is conducted in BF16 precision with gradient checkpointing enabled. We use the AdamW optimizer [17] with a cosine learning rate schedule, a peak learning rate of 2×10^{-5} , and a linear warmup over the first 5% of training steps. The per-device batch size is 4 with gradient accumulation of 2 steps, yielding an effective global batch size of 16.

We evaluate on two benchmarks. For in-domain evaluation, we use the held-out test split of WenetSpeech-Formal, comprising 31,932 samples, which shares the same domain distribution as the training data. For cross-domain evaluation, we use the Speechio-Formal test set [10], comprising 43,178 samples across 27 domain-specific subsets (ZH00000–ZH00026), including lecture recordings, interviews, meeting transcripts, and spontaneous dialogue.

We report three complementary metrics: CER (Character Error Rate, ↓), which measures character-level edit distance and captures surface-level transcription accuracy; ROUGE-L (↑), which reflects content preservation via longest common subsequence overlap; and BERTScore F1 (↑), which measures semantic similarity using contextual embeddings and is robust to paraphrase and minor wording differences. Together, these metrics assess both surface-level accuracy and semantic fidelity of the spoken-to-formal conversion.

We compare against two verbatim baselines: Qwen3-ASR-0.6B and Qwen3-ASR-1.7B without any fine-tuning, which represent the lower bound for spoken-to-formal quality and isolate the contribution of SFT training; and Whisper large-v3 [1], a widely-used open-source multilingual ASR model evaluated in verbatim mode as a cross-system reference.

5.2. Main Results

Table 4 reports spoken-to-formal ASR results on WenetSpeech-Formal and Speechio-Formal. FormalASR consistently outperforms the verbatim baselines across all metrics and both benchmarks, while Whisper large-v3 lags behind all Qwen3-ASR variants due to its verbatim transcription design.

Metric improvements. On WenetSpeech-Formal, our FormalASR-0.6B reduces CER from 0.2581 to 0.1770—a 31.4% relative reduction—while FormalASR-1.7B reduces CER from 0.2460 to 0.1606, a 34.7% relative reduction. ROUGE-L and BERTScore improve in tandem, confirming that the gains are not merely an artifact of shorter outputs: FormalASR-1.7B raises ROUGE-L from 0.8571 to 0.8896 and BERTScore F1 from 0.9268 to 0.9439 on WenetSpeech-Formal. These results demonstrate that the model simultaneously removes disfluencies and preserves semantic content.

Scale effect. FormalASR-1.7B consistently outperforms FormalASR-0.6B on both benchmarks across all three metrics, suggesting that a larger language model decoder provides stronger capacity for linguistic formalization. The advantage is most visible on BERTScore, where the 1.7B model scores 0.9439 versus 0.9359 for the 0.6B model on WenetSpeech-Formal and 0.9533 versus 0.9481 on Speechio-Formal.

Cross-domain generalization. On the cross-domain Speechio-Formal benchmark, both models maintain strong performance across 27 domain-specific subsets unseen during training. FormalASR-0.6B achieves a 28.8% relative CER reduction from 0.2252 to 0.1603, and FormalASR-1.7B achieves a 37.4% reduction from 0.2393 to 0.1499, demonstrating that the spoken-to-formal capability learned from WenetSpeech-Formal transfers broadly to diverse speech domains.

5.3. Inference Efficiency

A side benefit of spoken-to-formal conversion is that removing filler words and disfluencies shortens the output sequence, which directly reduces the number of autoregressive decoding steps. Figure 2 quantifies this effect for FormalASR-1.7B versus the verbatim Qwen3-ASR-1.7B baseline.

Output token reduction. The left panel of Figure 2 shows average output token counts on both benchmarks. On

Table 5. GGUF Quantization results of FormalASR on WenetSpeech-Formal. The “Sample Output” column shows model output for the same test utterance (spoken: “整个整个记整体的给您做个报表吧”; reference: “整体给您做个报表吧。”) across quantization levels.

Model	Precision	Model Size	CER ↓	ROUGE-L ↑	BERTScore ↑	Sample Output
FormalASR-0.6B	BF16	1.46 GB	0.1770	0.8769	0.9359	整个整体地给您做个报表吧。
	Q8_0	0.78 GB	0.1775	0.8766	0.9357	整个整体地给您做个报表吧。
	Q4_K	0.42 GB	0.1969	0.8627	0.9281	整个整体地给您做个报表吧。
FormalASR-1.7B	BF16	3.80 GB	0.1606	0.8896	0.9439	整体给您做个报表吧。
	Q8_0	2.03 GB	0.1607	0.8896	0.9438	整体给您做个报表吧。
	Q4_K	1.08 GB	0.1744	0.8805	0.9392	整体给您做个报表吧。

WenetSpeech-Formal, FormalASR-1.7B produces 14.3 tokens per utterance on average, compared to 18.5 for Qwen3-ASR-1.7B—a 22.8% reduction. On the cross-domain Speechio-Formal benchmark, the reduction is 14.3% from 18.5 to 15.8 tokens, confirming that the effect is consistent across domains.

Latency scaling. The right panel of Figure 2 plots per-sample decoding latency as a function of verbatim sentence length, grouped into five bins. For short utterances of 0–9 tokens, FormalASR-1.7B decodes in approximately 1,188 ms versus 1,364 ms for Qwen3-ASR-1.7B, a gap of roughly 176 ms. As sentence length grows, the advantage of FormalASR-1.7B becomes increasingly pronounced: in the 20–29 token bin, FormalASR-1.7B reduces latency by approximately 324 ms relative to Qwen3-ASR-1.7B, and in the longest bin of 40–49 tokens the gap widens to roughly 388 ms, with Qwen3-ASR-1.7B reaching approximately 3,292 ms while FormalASR-1.7B stays at approximately 2,904 ms. This super-linear scaling of the latency benefit with utterance length arises because longer verbatim transcripts tend to contain more disfluencies, so spoken-to-formal conversion removes proportionally more tokens and yields a larger reduction in decoding steps. The result makes FormalASR particularly advantageous for long-form speech such as meeting transcription and lecture recording.

5.4. Quantization

To further validate the on-device motivation, we evaluate post-training quantization on FormalASR checkpoints using the GGUF format [18], which is widely supported by on-device inference runtimes such as llama.cpp. We quantize both FormalASR-0.6B and FormalASR-1.7B to 8-bit (Q8_0) and 4-bit (Q4_K) precision and measure the resulting accuracy and model-size trade-offs on the WenetSpeech-Formal test set.

Table 5 reports the accuracy and model-size trade-offs under GGUF quantization, with a qualitative sample output column to illustrate output-level behavior across precision levels.

8-bit quantization is near-lossless. For FormalASR-1.7B, Q8_0 reduces model size by 47% from 3.80 GB to

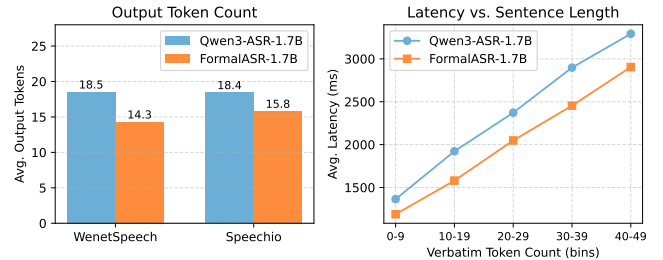


Fig. 2. Inference efficiency of FormalASR-1.7B compared with Qwen3-ASR-1.7B (verbatim). **Left:** average output token counts on WenetSpeech-Formal (18.5 for Qwen3-ASR-1.7B, 14.3 for FormalASR-1.7B) and Speechio-Formal (18.5 for Qwen3-ASR-1.7B, 15.8 for FormalASR-1.7B), showing that spoken-to-formal conversion consistently produces shorter sequences. **Right:** per-sample latency as a function of verbatim sentence length (token count bins); FormalASR-1.7B achieves lower latency across all sentence lengths, with the gap widening for longer utterances.

2.03 GB while incurring virtually no quality degradation: CER increases by only 0.0001, a 0.06% relative change, and ROUGE-L and BERTScore are unchanged to four decimal places. FormalASR-0.6B shows similarly negligible degradation under Q8_0: CER rises by just 0.0005 from 0.1770 to 0.1775, BERTScore drops by 0.0002, and memory shrinks by 47% from 1.46 GB to 0.78 GB. These results confirm that 8-bit GGUF quantization is a reliable compression strategy for FormalASR, preserving spoken-to-formal quality at nearly half the memory cost.

4-bit quantization offers strong compression with moderate degradation. Q4_K reduces model size by approximately 72% relative to BF16, from 3.80 GB to 1.08 GB for FormalASR-1.7B and from 1.46 GB to 0.42 GB for FormalASR-0.6B, enabling deployment on memory-constrained devices. The quality cost is moderate: FormalASR-1.7B Q4_K raises CER by 8.6% relative from 0.1606 to 0.1744 and lowers BERTScore by 0.5% from 0.9439 to 0.9392, while FormalASR-0.6B Q4_K shows a larger relative CER increase of 11.2% from 0.1770 to 0.1969. Notably, the 1.08 GB

FormalASR-1.7B Q4_K model still outperforms the 1.46 GB FormalASR-0.6B BF16 model across all three metrics, suggesting that the 1.7B model retains a quality advantage over the smaller model even after aggressive 4-bit compression.

Qualitative output analysis. The “Sample Output” column provides a concrete illustration of quantization behavior on a single test utterance. For FormalASR-1.7B, all three precision levels (BF16, Q8.0, Q4_K) produce the identical output “整体给您做个报表吧。”, which exactly matches the formal reference, demonstrating that the 1.7B model’s spoken-to-formal capability is fully preserved under quantization. For FormalASR-0.6B, all three precision levels consistently output “整个整体地给您做个报表吧。”, retaining the residual disfluency “整个” from the spoken input—a limitation of the smaller model that is unaffected by quantization precision. This pattern suggests that quantization does not introduce new formalization errors; rather, the output quality ceiling is determined by model capacity, not numerical precision.

6. CONCLUSION

We presented two contributions toward end-to-end spoken-to-formal Chinese ASR. First, we constructed and open-sourced WenetSpeech-Formal with 969K training samples and Speechio-Formal with 43K cross-domain test samples, two large-scale spoken-to-formal datasets built by rewriting verbatim transcriptions with DeepSeek-V3.2 and applying quality filtering, providing the first large-scale supervision resource for this task. Second, we fine-tuned FormalASR, compact end-to-end models at the 0.6B and 1.7B scales that directly transcribe spoken Chinese into formal written text without any auxiliary LLM at inference time. FormalASR achieves up to 37.4% relative CER reduction over verbatim baselines with consistent ROUGE-L and BERTScore gains across in-domain and cross-domain benchmarks, while also reducing decoding latency through shorter output sequences. GGUF quantization confirms practical deployability: Q8.0 is near-lossless at 47% reduced memory footprint, and Q4_K reduces model size by $\sim 72\%$ with moderate quality trade-off. Future work includes multilingual extension, RLHF-based formality optimization, and streaming inference for real-time transcription.

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Table 6. Bitsandbytes quantization results of FormalASR on WenetSpeech-Formal test set.

Model	Precision	Model Size	CER ↓	ROUGE-L ↑	BERTScore ↑
FormalASR-0.6B	BF16	~1.2 GB	0.1770	0.8769	0.9359
	INT8	~0.6 GB	0.1780	0.8761	0.9355
	INT4	~0.3 GB	0.3750	0.7582	0.8867
FormalASR-1.7B	BF16	~3.4 GB	0.1606	0.8896	0.9439
	INT8	~1.7 GB	0.1620	0.8887	0.9435
	INT4	~0.85 GB	0.2791	0.8104	0.9114

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A. APPENDIX

A.1. Bitsandbytes Quantization Results

Table 6 reports bitsandbytes [19] INT8/INT4 quantization results as a complement to the GGUF results in Section 5. INT8 is near-lossless (CER +0.0010 for 0.6B; +0.0014 for 1.7B) and halves memory, consistent with GGUF Q8_0. INT4, however, causes severe quality collapse: CER rises to 0.3750 for FormalASR-0.6B (+112% relative) and 0.2791 for FormalASR-1.7B (+74% relative), far worse than GGUF Q4_K (+11% and +9% respectively) at the same memory footprint. The gap stems from bitsandbytes’ uniform abs-max quantization versus GGUF’s per-block mixed-precision k-quants; GGUF Q4_K is therefore the recommended choice when aggressive compression is required.

A.2. Effect of GRPO

We explored GRPO [20] on top of the SFT checkpoint for the 1.7B model, using a formality reward (edit-distance reduction relative to the verbatim input) and a semantic fidelity reward (BERTScore F1 against the formal reference). As shown in Table 7, SFT+GRPO achieves CER 0.1609, ROUGE-L 0.8895, and BERTScore 0.9438, which is virtually identical to SFT alone at CER 0.1606, ROUGE-L 0.8896, and BERTScore 0.9439, indicating that the dense SFT supervision already saturates the reward landscape and leaves no room for policy improvement. We therefore adopt SFT as the final training strategy for FormalASR.

Table 7. Ablation on training strategy (1.7B model, WenetSpeech-Formal test set).

Configuration	CER ↓	ROUGE-L ↑	BERTScore ↑
No fine-tuning	0.2460	0.8571	0.9268
SFT only	0.1606	0.8896	0.9439
SFT + GRPO	0.1609	0.8895	0.9438